

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

1 Introduction

Definition 1.1. A **population** is a well-defined collection of objects.

Definition 1.2. A **census** is a collection of all desired information for all objects in the population.

Definition 1.3. A **sample** is a subset of population.

Definition 1.4. A **data** (set) is a collection of facts.

- A data set is usually represented by a set in the shape of $\{x_i\}_{i=1}^n$, where n is the number of observations in the data - namely **sample size** in the case of sample data - and $x_1, x_2, \dots, x_n$ represent individual observations in the data, where the subscripts are not necessarily related to any order among the observations.

Definition 1.5.0. A **variable** is a characteristic whose value may change from one object to another in the population.

Definition 1.5.1. A **categorical variable** is a variable whose possible values are categories.

Definition 1.5.2. A **numerical variable** is a variable whose possible values are numbers.

- A variable is usually represented by a letter (e.g., x or y).

Definition 1.6.1. A numerical variable is **discrete** if its set of possible values either is finite or else can be listed in an infinite sequence.

- In other words, a discrete numerical variable has a countable set of possible values.
- A special type of discrete variable is called **dummy variable**, which takes a binary value (mostly - if not always - 0 or 1) and represents the absence/presence of some categorical characteristic.

Definition 1.6.2. A numerical variable is **continuous** if its set of possible values consists of an entire interval on the number line.

- Sometimes a discrete variable can be modeled as continuous, i.e., treated as a continuous variable in practice. Such a treatment is more likely to be applied to relatively dense data.

Definition 1.7. A data (set) is said to be **univariate** (**bivariate**/**multivariate**) if it consists of observations on a single (each of two/more than two) variable(s).

- Descriptive vs inferential statistics:
 - Methods from descriptive statistics summarize and describe important features of (sample) data.
 - By contrast, methods from inferential statistics use sample information to draw some type of conclusion (make an inference of some sort) about the population.

2 Visual Methods of Descriptive Statistics

Definition 2.1.1. With a given data set, the **frequency** of a particular value of a discrete variable in the data set is the number of times that value occurs in the data set.

Definition 2.1.2. With a given data set, the **relative frequency** of a particular value of a discrete variable is the fraction or proportion of times the value occurs in the data set - i.e., the frequency divided by the number of observations in the data set.

- By definition,

$$relative\ frequency = \frac{frequency}{\#\ of\ observations}.$$

Method 2.2. To draw a **histogram for a discrete data** x (a discrete variable x of observations in the data set), we

1. calculate the frequency and relative frequency of each possible value of x ,
 2. mark all possible values of x on the horizontal axis, and
 3. above each possible value of x , draw a rectangle so that such rectangles are all (1) with heights that are all the frequencies (or are all the relative frequencies) of the corresponding value of x and (2) with equal widths.
- The above method ensures that the area of each rectangle is proportional to the relative frequency of the value.

Definition 2.3. **Class intervals** (classes) are a set of intervals that divide all possible values of a (continuous) variable so that (the value of) each observation falls in exactly one of the intervals.

- In cases where observations may occasionally fall on boundaries of class intervals, we may address this concern by using decimal numbers as boundaries (e.g., age, height, weight, etc.).

Method 2.4. To draw a **histogram with equal class widths for a continuous data** x , we

1. determine class intervals of the data,
2. calculate the frequency and relative frequency of each class interval,
3. mark the class boundaries on the horizontal axis, and
4. above each class interval, draw a rectangle so that such rectangles are all with heights that are all the frequencies (or are all the relative frequencies) of the corresponding class interval.

- A rule of thumb for relatively small sample sizes is

$$\# \text{ of classes} = \sqrt{\# \text{ of observations}}.$$

- When observations are relatively unevenly distributed, histograms with equal class widths may be rather uninformative and thus undesirable.

Method 2.5. To draw a **histogram with unequal class widths for a continuous data** x , we

1. determine class intervals of the data,
 2. calculate the frequency and relative frequency of each class interval,
 3. mark the class boundaries on the horizontal axis, and
 4. above each class interval, draw a rectangle so that for each rectangle, its height equals the relative frequency of the class divided by the width of the class.
- The resulting heights of the rectangles are usually known as densities.
 - The area of each rectangle equals the relative frequency of the corresponding class.
 - The unit of the class width is usually picked as the minimum value among the widths of all classes.

Definition 2.6. A histogram is said to be **unimodal (bimodal/multimodal)** if it has a unique (two/more than two) peak(s).

Definition 2.7. A histogram is said to be

1. **symmetric** if its right half is a mirror image of its left half;
2. **positively skewed** if the right or upper tail is stretched out compared with the left or lower tail;
3. **negatively skewed** if its stretching is to the left.